AI DEVOPS

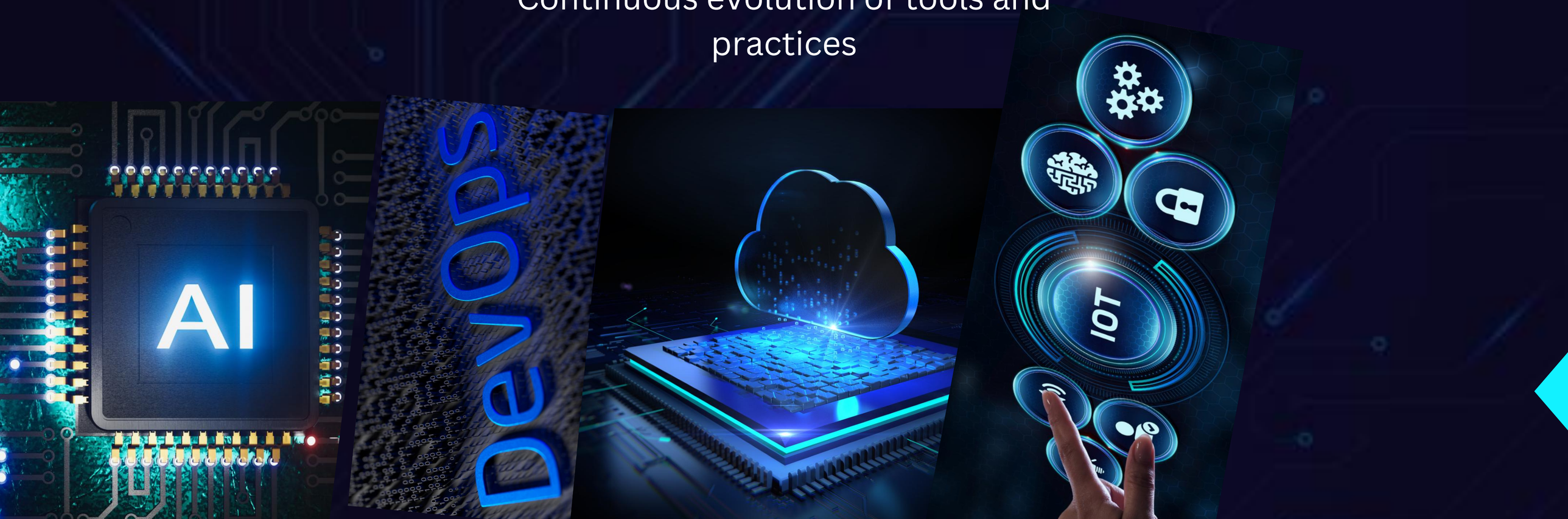
Exploring Cutting-Edge Technology Shaping The Future

AGENDA

- Introduction to Technology Trends
- Artificial Intelligence (AI) Overview
- DevOps Overview
- AI + DevOps: The Future
- Industry Use Cases

WHAT IS A TECHNOLOGY TREND?

Continuous evolution of tools and practices



WHY TRENDING TECHNOLOGIES MATTER?

01 Business growth and innovation

02 Automation and productivity

03 Career opportunities

WHAT IS ARTIFICIAL INTELLIGENCE?

Simulation of human intelligence by machines

TYPES :

01 Narrow AI

Designed to perform specific tasks, like facial recognition or driving.

02 General AI

AI with human-like cognitive abilities, capable of learning and performing various tasks.

AI SUBFIELDS

AI is a broad field with many subfields

- - Machine Learning
- - NLP (Natural Language Processing)
- - Computer Vision
- - Robotics

AI SUBFIELDS

AI is a broad field with many subfields

- - Machine Learning

Machine Learning is a subset of AI that allows systems to learn from data and improve over time without being explicitly programmed.
Key Concepts:
Supervised Learning: Model learns from labeled data (e.g., spam detection).
Unsupervised Learning: Model finds patterns in unlabeled data (e.g., customer segmentation).
Reinforcement Learning: Model learns by trial and error through rewards (e.g., game playing, robotics).
Applications:
Fraud detection
Recommendation systems (like Netflix,
Amazon)
Predictive analytics in finance and healthcare

AI SUBFIELDS

AI is a broad field with many subfields

- - NLP (Natural Language Processing)

NLP helps machines understand, interpret, and generate human language.

Key Concepts:

- Tokenization: Breaking text into words/sentences.
- Sentiment Analysis: Determining emotion in text.
- Named Entity Recognition: Identifying names, dates, locations in text.
- Language Generation: Producing human-like text (e.g., ChatGPT).

Applications:

- Chatbots and virtual assistants
- Language translation (Google Translate)
- Voice recognition (Alexa, Siri)
- Email spam detection

AI SUBFIELDS

AI is a broad field with many subfields

• - Computer Vision

Computer Vision enables machines to interpret and process visual data like images and videos.

Key Concepts:

- Image Classification: Identifying objects in an image.
- Object Detection: Locating multiple objects within an image.
- Image Segmentation: Breaking down an image into parts.
- Facial Recognition: Identifying individuals.

Applications:

- Self-driving cars (lane detection, pedestrian recognition)
- Face ID in smartphones
- Medical imaging diagnostics (X-rays, MRIs)
- Quality inspection in manufacturing

AI SUBFIELDS

AI is a broad field with many subfields

• - Robotics

Robotics involves building intelligent machines that can perform tasks autonomously or semi-autonomously.

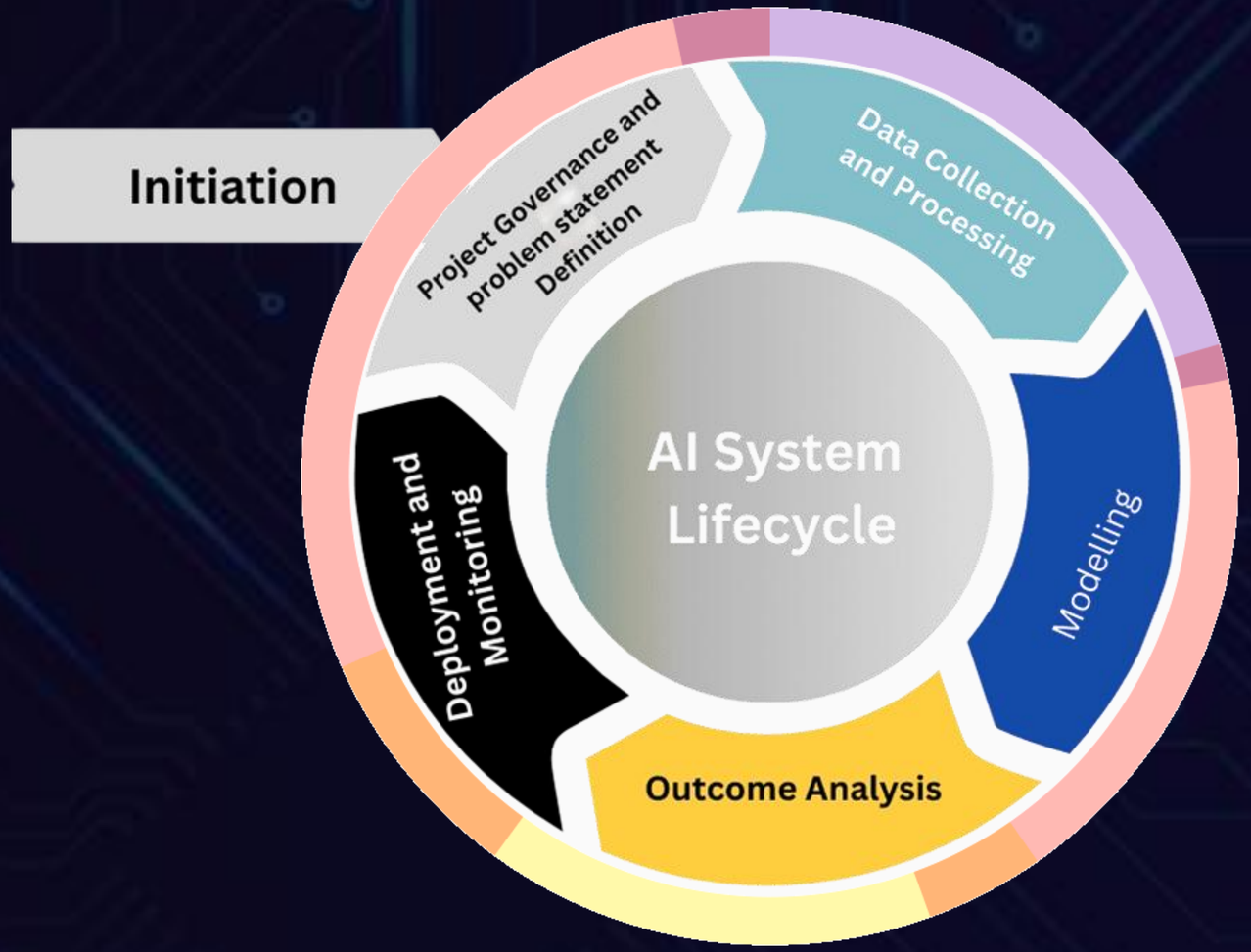
Key Concepts:

- Perception: Using sensors to gather data (e.g., cameras, LiDAR).
- Planning: Deciding how to move or act.
- Control: Executing planned actions smoothly and accurately.

Applications:

- Industrial robots (assembly lines)
- Delivery drones and autonomous vehicles
- Surgical robots
- Domestic robots (vacuum cleaners like Roomba)

AI LIFECYCLE



- Data Collection
→ Training
→ Evaluation
→ Deployment

AI APPLICATIONS

- •Healthcare,
- Finance,
- Retail,
- Transportation

AI APPLICATIONS

- Healthcare

Diagnosis & Imaging: AI models analyze medical images (like X-rays, MRIs) to detect diseases such as cancer or pneumonia faster and with high accuracy.

Personalized Medicine: AI recommends treatment plans based on genetic makeup, health history, and lifestyle.

Virtual Health Assistants: Chatbots and AI tools provide 24/7 support, medication reminders, and symptom checking (e.g., Ada, Babylon Health).

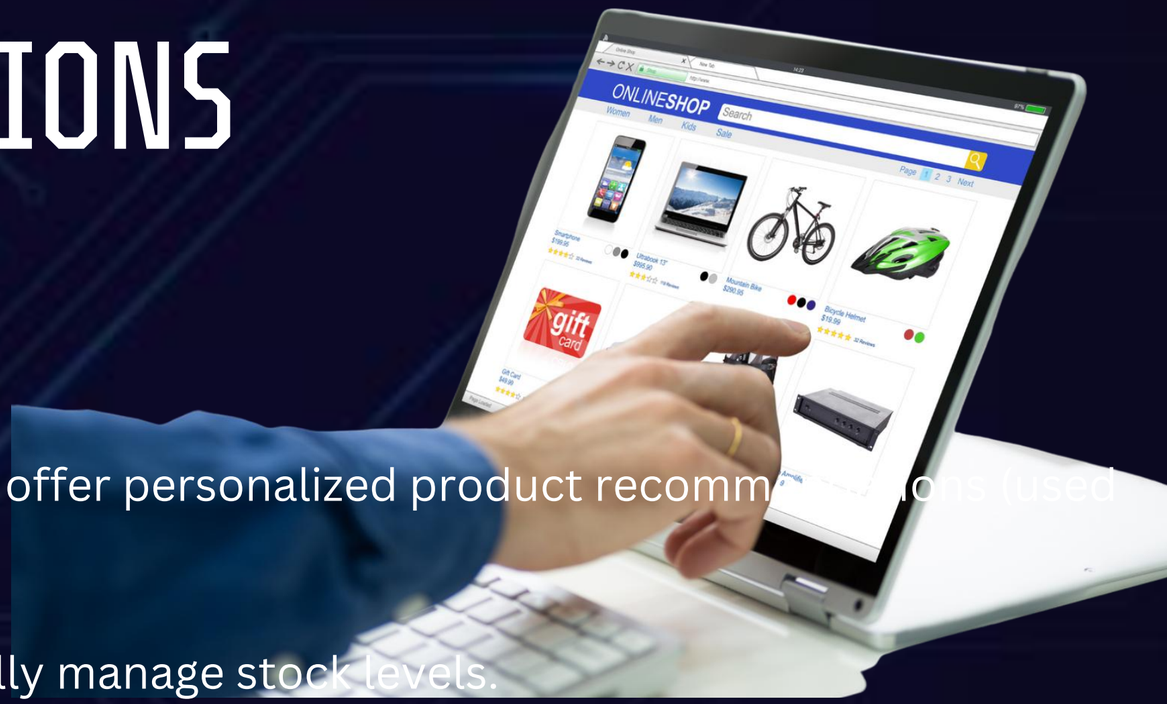
Drug Discovery: AI accelerates the process of finding new drugs by predicting molecule interactions and outcomes.

AI APPLICATIONS

- Finance

- Fraud Detection: AI analyzes transaction patterns in real time to detect suspicious activities.
- Algorithmic Trading: AI models make lightning-fast stock trades based on market data analysis.
- Credit Scoring: ML models help banks assess creditworthiness more accurately than traditional methods.
- Customer Service: Chatbots and AI-powered assistants (like in banking apps) provide 24/7 financial support.

AI APPLICATIONS



- Retail

- Customer Behavior Prediction: AI studies user behavior to offer personalized product recommendations (used by Amazon, Netflix).
- Inventory Management: Forecast demand and automatically manage stock levels.
- Visual Search: Customers upload pictures to find similar products online.
- AI-Powered Chatbots: Enhance customer service and guide shoppers through the buying process.

AI APPLICATIONS

- Transportation

- Self-Driving Cars: AI powers autonomous vehicles by analyzing surroundings through sensors and cameras (e.g., Tesla, Waymo).
- Traffic Management: AI predicts traffic congestion and adjusts signals accordingly (smart cities).
- Fleet Management: Optimize routes, fuel usage, and maintenance scheduling using predictive analytics.
- Public Transport: AI recommends optimal scheduling and predicts delays.

CASE STUDY: CHATGPT

✅ AI in NLP (Natural Language Processing)

What is NLP?

NLP stands for Natural Language Processing. It’s a branch of AI that helps machines understand, interpret, and generate human language.

How does ChatGPT use NLP?

ChatGPT reads and understands your text input (like a question or message), figures out what you mean, and responds in a human-like way.

🗣️ Example:

You type: “What is the capital of India?”

ChatGPT understands your question using NLP and replies: “New Delhi .”

CASE STUDY: CHATGPT

🔄 Real-time Conversational AI

- ChatGPT responds to your messages instantly, like chatting with a friend.
- It doesn’t just give pre-written answers — it generates them on the spot based on your question.

🗣️ Example:

You say: “Tell me a joke.”

ChatGPT creates a fresh joke instantly:

“Why don’t robots panic? Because they always keep their ‘compo-sure’.” 😄

CASE STUDY: CHATGPT

🌍 Applications in Customer Service, Education, etc.

Here's how ChatGPT is used in different areas:

1. Customer Service
   - Acts as a virtual assistant or chatbot.
   - Helps answer customer questions 24/7.
   - Example: A bank chatbot answering balance inquiries.
2. Education
   - Explains topics to students.
   - Helps with homework, summaries, or learning code.
   - Example: Explaining a math formula or writing an essay.
3. Healthcare (Basic Use)
   - Provides health information or reminders.
   - Example: "What are the symptoms of the flu?"
4. Business
   - Drafts emails, creates reports, or helps in documentation.
   - Saves time by automating repetitive writing tasks.

WHAT IS DEVOPS?

- Combination of Development & Operations

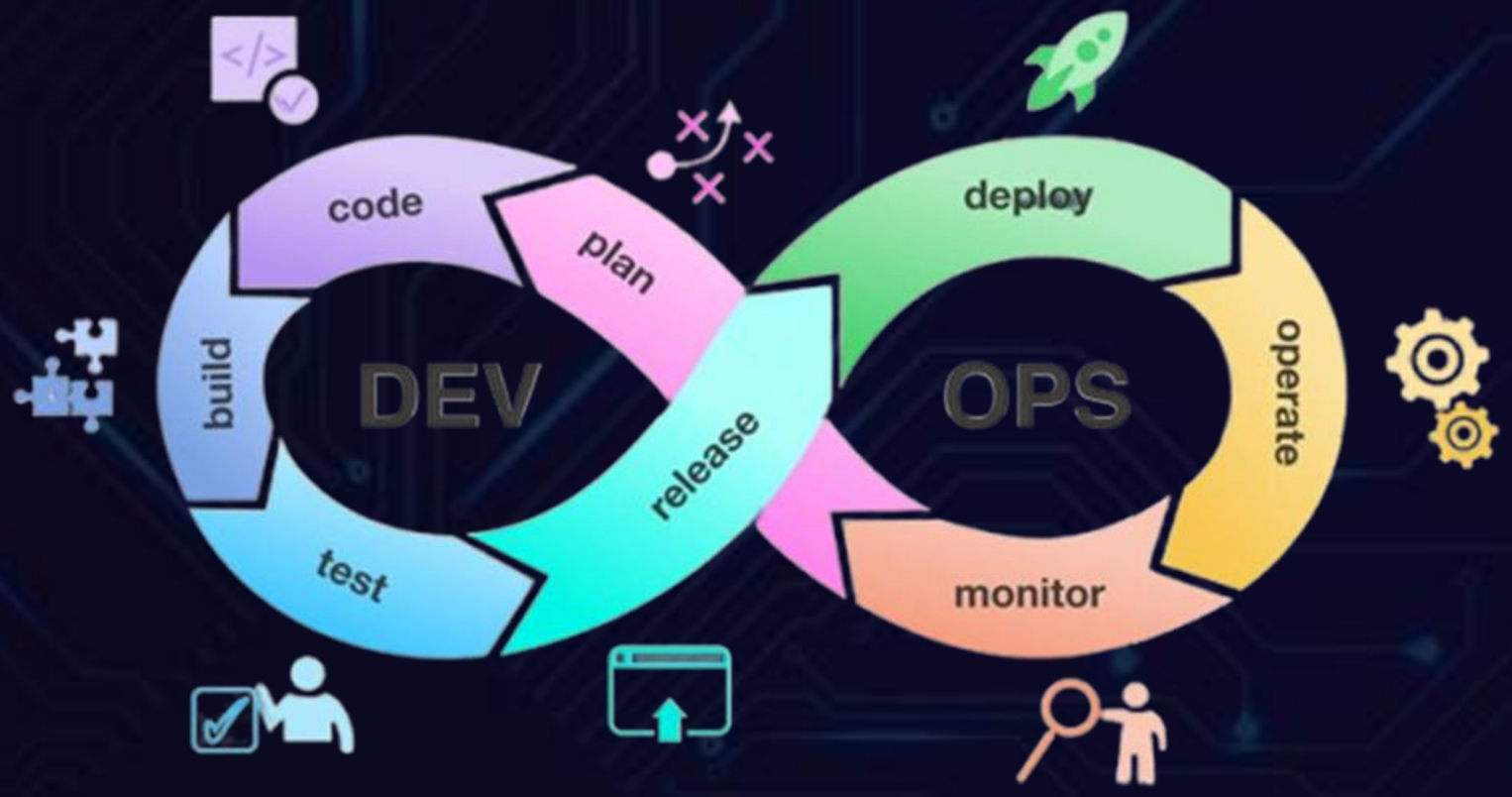
Before DevOps:
Developers write code ✅
Operations manage servers & deployments 🖥️ 🛠️
They worked separately, which caused delays and blame games ⛔
With DevOps:
Both teams collaborate 🤝
Everything is automated for speed and reliability ⚙️
Focus is on delivering software faster, better, and safer 🚀

- Culture of collaboration and automation

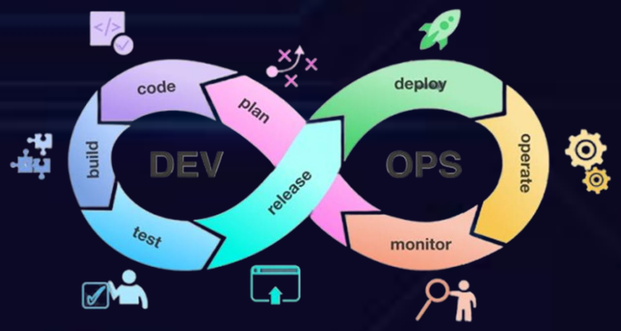
🤝 Culture of Automation
DevOps automates the boring & repetitive stuff like:
Code Testing 🧪 – done automatically (CI: Continuous Integration)
Deployment 🚀 – apps go live without manual effort (CD: Continuous Delivery)
Monitoring 📊 – systems are watched automatically to catch errors

🛠️🧪 Example
Let’s say you’re building an app like Swiggy.
With DevOps:
Code changes are tested & deployed automatically 🚀
New features go live daily or weekly, not monthly 📅🔁
Bugs are found and fixed faster 🔧
Everyone (devs + ops) works in sync 🤝🔄

DEVOPS LIFECYCLE

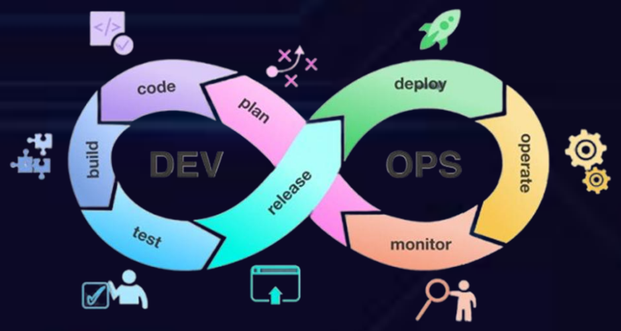


DEVOPS TOOLS



- Git,
- Jenkins,
- Docker,
- Kubernetes,
- Terraform,
- Prometheus

DEVOPS TOOLS



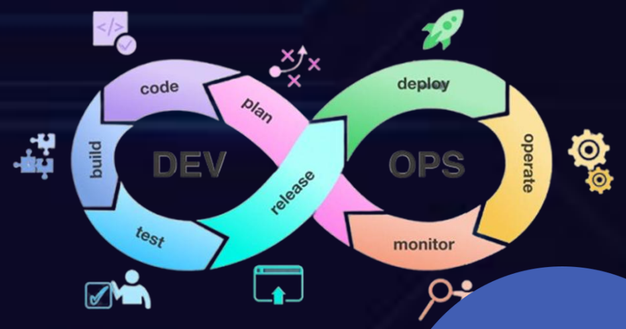
- Git

  Version Control System
  Keeps track of code changes.
  Allows multiple developers to collaborate on the same project.
  If something breaks, you can revert to an earlier version.
  📌 Example: “Undo” button for code!

- Jenkins

  Automation Server / CI Tool
  Automatically tests and builds your code whenever there’s a change.
  Helps implement Continuous Integration (CI).
  📌 Example: Every time a developer uploads code, Jenkins checks if it works!

DEVOPS TOOLS



- Docker

  Containerization Platform
  - Packages your app and its dependencies into one portable container.
  - Runs the same way on any system (Windows, Linux, cloud).

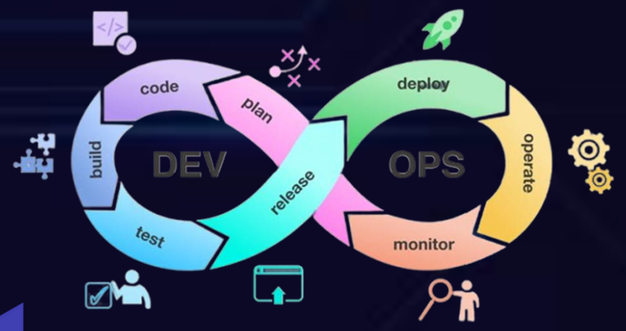
  📌 Example: Like packing all you need in a suitcase and taking it anywhere!

- Kubernetes

  Container Orchestration Tool
  - Manages, scales, and deploys many Docker containers.
  - Useful when your app becomes big and runs across many machines.

  📌 Example: Traffic police for containers — directs and manages them.

DEVOPS TOOLS



- Terraform

Infrastructure as Code (IaC) Tool

- Creates and manages cloud infrastructure (servers, databases, etc.) with code.
- Works with AWS, Azure, GCP, and more.

📌 Example: Instead of clicking on AWS console, write code to build cloud infrastructure.

- Prometheus

Monitoring Tool

- Tracks system performance like CPU, memory, uptime, etc.
- Alerts you when something goes wrong.

📌 Example: A health monitor for your servers and apps.

BENEFITS OF DEVOPS

- Faster delivery
- Improved collaboration
- Higher quality software

CI/CD PIPELINE

- Automates code integration, testing, and delivery

Continuous Integration (CI)

CI is the process of automatically integrating code changes from multiple developers into a shared repository several times a day.

✅ Key Features:

Automated builds and tests run every time code is pushed.

Detects bugs early in the development cycle.

Helps maintain a clean, working codebase at all times.

⚙ Example Tools:

GitHub Actions

Jenkins

CircleCI

📌 Example: When a developer pushes code to GitHub, the CI pipeline automatically runs unit tests to verify nothing is broken.

CI/CD PIPELINE

- Automates code integration, testing, and delivery

🚚 Continuous Delivery (CD)

CD ensures that the tested code is automatically delivered to a staging or production environment for release.

✅ Key Features:

- Code is always in a deployable state.
- Manual approval can be added before going live.
- Reduces risks of release-day failures.

📌 Example: After passing tests, the code is pushed to a staging server, ready for QA or final approval.

CI/CD PIPELINE

- Reduces errors and downtime

Early Bug Detection

- With automated testing in Continuous Integration (CI), code is tested every time it's committed.
- Issues are caught early, before they reach production.
- This reduces the risk of bugs piling up or going unnoticed.

📌 Example: A failing test during CI stops broken code from being deployed, preventing system crashes later.

Faster Recovery

CI/CD pipelines allow quick rollbacks or redeploys.

You can recover from failures faster, minimizing downtime.

📌 Example: If a deployment breaks something in production, you can roll back to the previous stable version instantly.

HOW AI ENHANCES DEVOPS

Predictive analytics

AI can analyze vast amounts of historical and real-time data from DevOps pipelines to predict potential issues before they occur.
✅ Benefits:
Forecast deployment failures, performance bottlenecks, or capacity issues.
Optimize resource allocation (like CPU/memory usage trends).
Plan infrastructure scaling before traffic spikes occur.
📌 Example: AI can predict that a service is likely to crash during high traffic based on past deployments and trigger autoscaling or preventive alerts.

Anomaly detection

AI models can continuously monitor logs, metrics, and traces to detect unusual patterns or behavior—much faster than humans or traditional scripts.
✅ Benefits:
Spot security threats (e.g., sudden access spikes).
Identify unexpected errors, latency, or crash loops.
Trigger alerts and even automated remediation.
📌 Example: AI notices that a container restarts 3x more than usual and notifies DevOps to investigate or automatically rolls it back.

HOW AI ENHANCES DEVOPS

Automated testing & monitoring

AI can optimize and automate testing in CI/CD pipelines and enhance monitoring post-deployment.

✅ Benefits:

- Test selection optimization: AI chooses the most relevant tests to run, reducing build times.
- Self-healing systems: AI can restart services or roll back code autonomously.
- Smart alerting: Reduce false positives and alert fatigue by learning normal vs. abnormal patterns.

📌 Example: During deployment, AI skips redundant tests and focuses on impacted modules—cutting test time in half.

INDUSTRY USE CASES

Amazon: AI in logistics, DevOps for deployment

AI in Logistics	DevOps in Deployment
Robot-powered warehouses	Automated testing & deployments
Demand prediction	Safe, gradual rollouts
Route optimization	Rapid updates with fewer errors
Smart inventory	Scalable, resilient infrastructure

INDUSTRY USE CASES

Netflix: AI recommendations, scalable CI/CD

AI in Netflix	DevOps in Netflix
Personalized content suggestions	Highly automated CI/CD pipelines
Dynamic, user-specific thumbnails	Microservices for agile development
Predictive content production	Chaos engineering for reliability
Adaptive streaming quality	Auto-scaling cloud infrastructure

AI CAREER PATHS

- AI Engineer,
- Data Scientist,
- ML Engineer,
- NLP Specialist

DEVOPS CAREER PATHS

- DevOps Engineer,
- SRE,
- Cloud Engineer

SKILLS NEEDED

- Programming,
- Cloud,
- Data,
- Automation,
- Collaboration

CERTIFICATIONS

- AI: Google,
- Microsoft,
- IBM

CERTIFICATIONS

- AI: Google,
- Microsoft,
- IBM

- DevOps: AWS, Docker, Kubernete

SUMMARY

AI and DevOps are shaping the future

Embrace continuous learning